



Part A: Conceptual Foundation

1. What is Regularization in Machine Learning? Why is it needed?

Regularization is a technique used to reduce overfitting in machine learning models by adding a penalty term to the loss function. It prevents the model from learning noise and overly complex patterns from training data. Regularization improves model generalization and helps achieve better performance on unseen data.



Regularization Concept

Training Data → Model Training → Overfitting ❌

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Apply Regularization

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Better Generalization ✅

2. Difference between Ridge Regression (L2) and Lasso Regression (L1)

Ridge Regression (L2) adds the squared values of coefficients as a penalty term, which reduces coefficient values but rarely makes them zero. Lasso Regression (L1) adds the absolute values of coefficients, which can shrink some coefficients completely to zero. Ridge is useful for reducing model complexity, while Lasso also performs feature selection.



Ridge vs Lasso

Feature	Ridge Regression (L2) Lasso Regression (L1)	
Penalty	Squared coefficients	Absolute coefficients
Formula	L2 penalty	L1 penalty
Feature Selection	❌ No	✅ Yes
Coefficients	Reduced but not zero	Some become zero
Main Use	Reduce overfitting	Reduce overfitting + feature selection



Visual Idea

Ridge : $5 \rightarrow 3 \rightarrow 1.5 \rightarrow 0.8$

Lasso : $5 \rightarrow 2 \rightarrow 0$

3. What is Cross-Validation and why is it important?

Cross-validation is a model evaluation technique where the dataset is divided into multiple parts to test model performance on different subsets. It helps estimate how well the model will perform on unseen data. Cross-validation improves reliability, reduces overfitting risk, and helps in selecting the best model and hyperparameters.

Cross-Validation Workflow

Dataset

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Split into folds

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Train on some folds

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Test on remaining fold

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Repeat process

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Average Performance Score

4. Explain the following cross-validation techniques:

• K-Fold Cross-Validation

In K-Fold Cross-Validation, the dataset is divided into K equal parts (folds). The model is trained on K-1 folds and tested on the remaining fold. This process repeats K times, and the average performance is calculated.

Example

Fold 1 → Test

Fold 2 → Test

Fold 3 → Test

Fold 4 → Test

Fold 5 → Test

- **Stratified K-Fold Cross-Validation**

Stratified K-Fold ensures that each fold maintains the same proportion of target variable distribution as the original dataset. It is useful when the dataset is imbalanced or target values are unevenly distributed.



Visual Representation

Original Dataset

70% Class A

30% Class B

Each Fold

70% Class A

30% Class B

- **Leave-One-Out Cross-Validation (LOOCV)**

LOOCV is a special case of K-Fold where K equals the number of samples. The model is trained on all samples except one and tested on the remaining sample. This process repeats for every data point.



LOOCV Concept

Train: All samples except 1

Test : Remaining 1 sample

Repeat for every sample

- **Time Series Split**

Time Series Split is used for sequential or time-dependent data. Training is performed on past data, and testing is done on future data while maintaining chronological order. It avoids data leakage in time-series problems.

Time Series Split

Past Data → Future Data

[Train Train Train] [Test]

5. Why are tree-based models less sensitive to feature scaling?

Tree-based models split data based on feature values and decision rules rather than distance calculations. Since they do not depend on magnitude or scale, normalization and standardization usually do not affect their performance. Therefore, feature scaling is generally unnecessary for tree-based algorithms like Decision Trees and Random Forests.

Tree-Based Model Concept

Feature Value > Threshold?

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Yes / No

Important Point

- Linear models and SVR usually require feature scaling.
 - Decision Trees and Random Forests generally do not require scaling.
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Conclusion

This section explained important machine learning concepts including regularization, Ridge and Lasso regression, cross-validation techniques, and tree-based models. These concepts